CoMind

MILESTONE 1

Concept Drafts • Comparison • Prompt Log • Project Approval

Team Members:

- **Marina Medhat**
- **Eman Youssef**
- **Ayat Mohamed**
- **Alaa Sameh**
- **Mohamed Kamal**

1. Project Idea for Approval

Project overview

CoMind is proposed as a web-based educational chatbot that helps Arabic-speaking beginners learn Egyptian Sign Language (ESL) through verified sign-language videos. Users enter a word or phrase in Arabic or English, and the system searches a predefined dataset for the corresponding sign video. When an exact verified match is unavailable, CoMind explains that it could not find a verified result rather than presenting an unverified demonstration as correct.

Target user group

Arabic-speaking beginners who want to learn common Egyptian Sign Language words and phrases independently using visual examples and short practice activities.

Problem statement

Beginners may have difficulty finding accessible, consistent learning materials for Egyptian Sign Language and checking whether a video corresponds to the word they searched for. CoMind aims to simplify the learning process by retrieving labeled videos from a predefined dataset and making the outcome of each search clear.

Core features proposed

- Word or phrase input in Arabic or English.
- Search for an exact match in a predefined sign-video dataset.
- Display the retrieved video with its word or phrase label.
- Clearly report when no verified match is available.
- Offer related entries only if they are present in the dataset and clearly label them as related, not exact matches.
- Provide a video-based multiple-choice quiz for practicing sign recognition.

Proposed distinction

The proposed value is the combination of transparent dataset-based video retrieval, explicit handling of unavailable signs, and video-based recognition practice in one beginner-focused experience. This is a proposed distinction, not a claim that no similar system exists. The team should confirm with the instructor that no other team is proposing the same idea for the same user group.

2. Selected HAI Principle: Explainability

Explainability means helping users understand what result the system returned and how to interpret it. CoMind applies this principle by labeling the retrieved sign video, distinguishing exact matches from any related suggestions, and clearly communicating when a verified result is unavailable. The system should not imply that an unverified or unrelated sign is correct.

3. Concept Draft A — ChatGPT

Concept title: Simple Sign-Learning Conversation

Core interaction flow

- The learner types a word or phrase, for example, “thank you” or “good morning.”
- CoMind searches the predefined sign-language dataset.
- If an exact verified match is found, the system displays the video with a clear label and a short explanation, where available.
- If no verified match is found, the system says so and does not generate an unverified sign demonstration.
- The learner can try another word or open a video-based quiz to practice.

Strengths

- Simple and beginner-friendly flow.
- Prioritizes visual learning.
- Clearly handles unavailable signs.
- Supports follow-up learning.

Weakness

The flow does not explicitly define how exact matches should be distinguished from related results.

4. Concept Draft B — Claude

Concept title: Transparent Sign Retrieval

Core interaction flow

- The learner enters a word or phrase.
- CoMind searches the dataset for an exact match first.
- If an exact verified match exists, the system displays the corresponding sign video.
- If no exact match exists, the system may offer related dataset entries only when suitable entries exist, and labels them clearly as related suggestions.
- If no suitable verified entry exists, the system explains that it could not find the requested sign.
- The learner can select a clearly labeled related entry or search for another word.

Strengths

- Makes result types explicit.
- Improves transparency around exact and related matches.
- Defines a structured no-match path.
- Supports alternative searches without claiming an unrelated sign is exact.

Weakness

Related suggestions could confuse beginners unless the interface labels them clearly and their relationship is verified.

5. Side-by-Side Comparison

Criterion	Draft A — ChatGPT	Draft B — Claude
Interaction style	Simple, conversational	Structured, result-type focused
Beginner friendliness	Strong	Strong
Unavailable result	Clearly reports no match	Clearly reports no suitable match
Exact vs. related entries	Less explicit	Explicit distinction
Main strength	Simplicity	Transparency
Main weakness	Related results are underspecified	Related suggestions require careful labeling

Chosen direction

The team proposes a hybrid direction: keep Draft A’s simple beginner-friendly interaction and adopt Draft B’s explicit distinction between exact matches and related suggestions. The system will prioritize verified exact matches, label any related entries clearly, and state when no verified result is available. This direction supports Explainability without making the interaction unnecessarily complex.

6. Appendix A — AI Prompt Log

Entry A — ChatGPT (draft prompt; verify against chat history)

Design a concept draft for the core interaction flow of a web-based educational chatbot that helps Arabic-speaking beginners learn Egyptian Sign Language (ESL). The user should be able to enter a word or phrase, and the chatbot should retrieve the corresponding sign video from a predefined dataset. If no verified match is available, the chatbot should clearly communicate this instead of generating an unverified sign demonstration. Describe the interaction step by step, including user input, search, video display, no-match handling, and follow-up interactions. Focus on Explainability and keep the concept suitable for beginners. Do not assume features that are not specified.

Tool: ChatGPT • **What it gave:** A simple interaction flow based on user input, dataset search, video display, unavailable-result handling, and follow-up learning.

What the team changed and why: The team retained the simple flow and proposed adding a clearer distinction between exact matches and related entries after comparing the concepts.

Entry B — Claude (draft prompt; verify against chat history)

Design a concept draft for the core interaction flow of a web-based educational chatbot that helps Arabic-speaking beginners learn Egyptian Sign Language (ESL). The user enters a word or phrase, and the chatbot searches a predefined dataset for the corresponding sign video. Prioritize exact verified matches; if no exact match exists, describe how clearly labeled related entries could be offered only when appropriate, and how the system should report when no suitable result exists. Explain each interaction step and focus on transparency and Explainability. Keep the experience simple for beginners and do not assume unsupported capabilities.

Tool: Claude • **What it gave:** A structured retrieval flow that distinguishes exact matches, related entries, and unavailable results.

What the team changed and why: The team retained the explicit result-type distinction while simplifying the interaction to keep it beginner-friendly.

Other prompts used during prototype work

For every additional AI prompt used to generate or revise the interface, HTML, quiz, or other artifacts, add the exact prompt (verbatim), the tool, the output received, and the team's change with its reason. Do not label a paraphrase as an exact prompt.

7. Prototype and Live Demo Plan

ISHARA | Learn Sign Language